

Cleft Palate, Proliferative Retinopathy, and Developmental Delay (MONDO:0033641): A Comprehensive Disease Report

Summary

Cleft palate, proliferative retinopathy, and developmental delay (MONDO:0033641) is an **ultra-rare autosomal-recessive Mendelian syndrome** caused by **biallelic loss-of-function variants in *LRRC32*** (encoding **GARP**, Glycoprotein A Repetitions Predominant; chromosome 11q13.5). To date, the disorder has been reported in only approximately **four patients from consanguineous families**, making it one of the least-characterized Mendelian conditions in the medical literature. The disease was first delineated by Harel and colleagues in 2019, who identified a homozygous stop-gain variant (*LRRC32* c.1630C>T; p.(Arg544Ter)) segregating in two consanguineous families with the defining triad of developmental delay, cleft palate, and proliferative retinopathy. A second, independent family was reported in 2022 with a distinct homozygous missense variant, confirming the gene–disease association and expanding the phenotype to include severe pre- and postnatal growth retardation and dysmorphic features.

The **mechanistic basis** of the syndrome is loss of GARP function. GARP is a cell-surface docking receptor that binds and presents the **latent form of transforming growth factor-beta (TGF- β)** on the plasma membrane, where it is subsequently activated in an integrin-dependent manner (chiefly via $\alpha v\beta 8/\alpha v\beta 6$) to release mature, signaling-competent TGF- β . Loss of GARP therefore diminishes **latent TGF- β presentation and activation**, reducing downstream **SMAD2/3 signaling**. This mechanism has been directly demonstrated in the developing secondary palate: *Garp/Lrrc32*-null mice show reduced SMAD2 phosphorylation and failed apoptosis in the palatal medial edge epithelium (MEE), causing cleft palate and phenocopying *Tgfb3*-null mice. Because the TGF- β pathway is also essential for retinal and central nervous system development, the retinopathy and neurodevelopmental features are attributed — largely by inference — to the same signaling deficit operating in those tissues.

There is **no targeted or disease-modifying therapy**. Management is entirely **supportive and multidisciplinary**: surgical palatoplasty for the cleft palate, vitreoretinal surgery/ophthalmologic management for the proliferative retinopathy, and developmental/rehabilitative support for the neurodevelopmental delay, combined with genetic counseling (25% sibling recurrence risk in an autosomal-recessive pedigree). Because of the extreme rarity — no new cases have been described since 2022 — much of the clinical natural history, epidemiology, prognosis, and treatment response data remain undefined, and this report explicitly flags those gaps.

Key Findings

Finding 1 — *LRRC32* (GARP) biallelic loss-of-function is the cause (autosomal recessive)

The syndrome is caused by **biallelic (homozygous) loss-of-function variants in *LRRC32***, transmitted in an **autosomal-recessive** pattern. Harel et al. (2019) identified a **homozygous stop-gain variant, *LRRC32* c.1630C>T; p.(Arg544Ter)**, segregating in **two consanguineous families comprising three affected individuals**, all presenting with the triad of developmental delay, cleft palate, and proliferative retinopathy. The original report states directly: *"We identified a homozygous stop-gain variant in LRRC32 (c.1630C>T; p.(Arg544Ter)) in two families with developmental delay, cleft palate, and proliferative retinopathy"* (PMID: [30976112](#)).

A **second, independent line of evidence** came from Hexner-Erlichman et al. (2022), who used whole-exome sequencing to identify a **distinct homozygous missense variant** — a substitution of a highly conserved isoleucine to threonine — in a **fourth patient** from another consanguineous family: *"Whole exome sequencing (WES) revealed a very rare homozygous missense variant in the LRRC32 gene, which resulted in substitution of a highly conserved isoleucine to threonine"* (PMID: [35656379](#)). This second family, harboring a different variant type (missense vs. nonsense) yet producing an overlapping phenotype, provides **allelic heterogeneity** that reinforces the causal gene–disease relationship. Both families were consanguineous, consistent with a rare recessive disorder enriched in inbred pedigrees.

Evidence source types: human clinical (two independent reports); genetic (WES/segregation).

Finding 2 — Mechanism: GARP loss impairs latent TGF- β 3 presentation/activation, reducing SMAD2 signaling and apoptosis in palatal medial edge epithelium

The molecular mechanism was established in a mouse model by Wu et al. (2017). ***Garp/Lrrc32*-null mice die within 24 hours of birth with an isolated failure of secondary palate fusion.** At embryonic day 14.5 (E14.5), the palatal **medial edge epithelial (MEE)** cells show **decreased apoptosis** and **reduced SMAD2 phosphorylation**: *"we observed decreased apoptosis and SMAD2 phosphorylation in the medial edge epithelial cells of the palatal shelf of GARP KO embryos at embryonic day 14.5"* (PMID: 28912269).

GARP and TGF β 3 co-localize in MEE cells, physically interact, and GARP is required for cell-surface display of membrane-associated latent TGF β 3: *"GARP is indispensable for the surface expression of membrane-associated latent TGF β 3"* (PMID: 28912269). Critically, the GARP-null palatal defect **phenocopies the *Tgfb3*-null mouse**: *"the failure to develop the secondary palate and concurrent reduction of SMAD phosphorylation without other defects in GARP KO mice phenocopied TGF β 3 KO mice"* (PMID: 28912269). This places GARP squarely upstream of TGF- β 3/SMAD signaling in palatogenesis and provides the direct mechanistic explanation for the cleft-palate component of the human syndrome.

More broadly, GARP presents latent TGF- β 1 on the surface of regulatory T cells and platelets, where activation requires **integrin α v β 8** and release (or, per newer models, allosteric exposure) of mature TGF- β — a mechanism corroborated by multiple structural and immunological studies (see Evidence Base).

Evidence source types: model organism (mouse knockout); in vitro (co-localization, interaction, SMAD phosphorylation assays).

Finding 3 — *LRRC32* is recessive-compatible in gnomAD; the recurrent pathogenic allele p.(Arg544Ter) is ultra-rare with no homozygotes

Population-genetic analysis of gnomAD is fully consistent with a rare recessive disease model. For *LRRC32* (ENSG00000137507, GRCh38 chr11:76,657,524–76,670,747), gnomAD v4 reports **pLI = 0.66** and an **observed/expected loss-of-function ratio (oe_lof) = 0.33** (90% CI 0.17–0.70), with **5 observed vs. 15.0 expected LoF alleles** (lof_z = 2.19). These values indicate that the gene tolerates heterozygous LoF variation (i.e., carriers are viable and present in the population), exactly as expected for a recessive disorder where only biallelic loss is pathogenic.

The recurrent disease allele **c.1630C>T p.(Arg544Ter) (rs369867819)** is **ultra-rare**, with an exome allele frequency $\approx 1.37 \times 10^{-6}$ (2 heterozygous alleles observed) and **no homozygotes** in gnomAD. A missense change at the same codon, p.Arg544Gln, is also seen at very low frequency (AF $\approx 2.05 \times 10^{-6}$). The absence of homozygotes in a reference database of >700,000 individuals is precisely what is expected for an ultra-rare, severe recessive condition and adds population-level support to the pathogenicity of the reported allele.

Evidence source types: computational/population-genetic (gnomAD v4 constraint metrics and allele frequencies).

Metric (gnomAD v4, <i>LRRC32</i>)	Value	Interpretation
pLI	0.66	Moderate LoF intolerance
oe_lof (90% CI)	0.33 (0.17–0.70)	Some constraint; recessive-compatible
Observed / Expected LoF	5 / 15.0	Fewer LoF than expected
lof_z	2.19	Mild constraint
p.(Arg544Ter) exome AF	$\approx 1.37 \times 10^{-6}$	Ultra-rare
p.(Arg544Ter) homozygotes	0	Consistent with severe recessive disease

Finding 4 — Clinical triad and TGF- β developmental basis; neonatal lethality in the mouse null limits modeling

The **defining clinical triad** is **cleft palate + proliferative retinopathy + developmental delay**, congenital in onset, reported in two consanguineous families (three individuals; Harel et al. 2019). The unifying developmental explanation is the **TGF- β pathway**, which is essential for both palatogenesis and retinal development: *"The transforming growth factor-beta (TGF β) signaling pathway is essential for palatogenesis and retinal development"* (PMID: 30976112).

A key limitation of the animal model is that **complete *Garp*-null mice die within 24 hours of birth**: *"Garp-null mice have palate defects and die within 24 h after birth"* (PMID: 30976112). This neonatal lethality means the mouse recapitulates the **palatal** defect but **cannot model the postnatal ocular (proliferative retinopathy) or neurodevelopmental (developmental delay)** features, which require postnatal survival to manifest and assess. The 2022 fourth patient additionally exhibited **severe pre- and postnatal growth retardation and dysmorphic features**, expanding the recognized phenotypic spectrum beyond the original triad.

Evidence source types: human clinical (phenotype); model organism (lethality/limitation).

Full Section-by-Section Report

1. Disease Information

Overview. MONDO:0033641 is an ultra-rare, congenital, autosomal-recessive multisystem syndrome defined by the co-occurrence of **cleft palate**, **proliferative retinopathy**, and **developmental delay**. It results from complete or near-complete loss of GARP (encoded by *LRRC32*), a chaperone/docking protein for latent TGF- β . The disorder was newly delineated as a Mendelian entity in 2019.

Key identifiers. - **MONDO:** MONDO:0033641 - **Gene:** *LRRC32* (HGNC:4161), also known as **GARP** - **OMIM:** The molecular entity is captured through the *LRRC32* gene entry (OMIM *137207); a discrete OMIM phenotype MIM number, if assigned, post-dates the original 2019 report. (Not definitively available.) - **Orphanet / ICD-10 / ICD-11 / MeSH:** No dedicated code was identified for this specific triad syndrome; component phenotypes map to generic terms (cleft palate ICD-10 Q35; congenital retinal disorders; unspecified developmental delay). (Not available as a unified code.)

Synonyms / alternative names. "LRRC32-related syndrome"; "GARP deficiency syndrome"; "cleft palate–proliferative retinopathy–developmental delay syndrome."

Data provenance. All information is derived from **aggregated disease-level and case-report literature** (two peer-reviewed clinical/genetic reports describing four patients) plus a mechanistic mouse study and public population databases — **not** from large EHR cohorts.

2. Etiology

Causal factors — genetic. The sole established cause is **biallelic loss-of-function of *LRRC32*/GARP**. Two variant classes are documented: a **nonsense** allele (p.Arg544Ter) and a **missense** allele (conserved Ile→Thr) ([PMID: 30976112](#); [PMID: 35656379](#)).

Genetic risk factors. **Consanguinity** is the principal risk factor, as expected for an ultra-rare recessive disorder; both reported families were consanguineous. Carrier status (heterozygosity) confers no known phenotype.

Environmental risk factors / protective factors / gene–environment interactions. Not available / not applicable. No environmental contributors, protective alleles, dietary factors, or gene–environment interactions have been described for this monogenic disorder. Given the deterministic Mendelian etiology, environmental modifiers are unlikely to be primary drivers, though they cannot be excluded as modifiers of severity.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset	Frequency (of ~4 reported)
Cleft palate	Physical malformation	HP:0000175	Congenital	Core triad feature
Proliferative retinopathy	Clinical/ ophthalmologic sign	HP:0000556 (retinopathy) / HP:0008046 (abnormal retinal vasculature)	Congenital/ early	Core triad feature
Developmental delay / intellectual disability	Neurodevelopmental	HP:0001263 (global developmental delay)	Infancy	Core triad feature
Severe pre-/ postnatal growth retardation	Growth abnormality	HP:0001511 / HP:0008897	Prenatal onset	4th patient (2022)
Dysmorphic facial features	Physical manifestation	HP:0001999	Congenital	4th patient (2022)

Characteristics. Onset is **congenital** for the triad. Severity appears **moderate to severe** and the course is best described as **stable/static** (a congenital malformation syndrome rather than a progressive degeneration), although the proliferative retinopathy component carries risk of progression to vision loss without intervention. **Frequency data are qualitative only**, given the tiny patient count.

Quality-of-life impact. Not formally measured (no EQ-5D/SF-36/PROMIS data). By clinical inference, cleft palate impairs feeding and speech; proliferative retinopathy threatens vision; developmental delay affects cognition, communication, and independence — collectively implying substantial lifelong QoL burden. (Quantitative data **not available**.)

4. Genetic / Molecular Information

- **Causal gene:** *LRRC32* (GARP), HGNC:4161, located at **11q13.5** (GRCh38 chr11:76,657,524–76,670,747; ENSG00000137507).
- **Pathogenic variants:**
- **c.1630C>T; p.(Arg544Ter)** — nonsense/stop-gain; rs369867819; germline; homozygous in affected individuals; predicted loss of function (truncation). gnomAD exome AF $\approx 1.37 \times 10^{-6}$, no homozygotes.
- **Homozygous missense, conserved Ile→Thr** — germline; function-impairing; reported 2022.
- **Variant classification:** Consistent with **pathogenic/likely pathogenic** per ACMG criteria (ultra-rare, homozygous in affected consanguineous families, segregation, LoF mechanism, functional support from mouse ortholog).
- **Somatic vs. germline:** **Germline** (constitutional, inherited).
- **Functional consequence: Loss of function** — reduced/absent GARP-mediated latent TGF- β presentation.
- **Modifier genes / epigenetics / chromosomal abnormalities: Not available.** No modifier loci, methylation changes, or large-scale cytogenetic rearrangements have been implicated; the disorder is a single-gene point-mutation disorder.

5. Environmental Information

Not applicable. No environmental factors, lifestyle contributors, toxins, or infectious agents are associated with this monogenic recessive syndrome. (No CTD/TOXNET or exposure associations identified.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic LoF mutation in *LRRC32*** (nonsense p.Arg544Ter or function-impairing missense) → **leads to** loss/severe reduction of functional GARP protein. (*Demonstrated: human genetics.*)
2. Loss of GARP → **results in** failure to display membrane-associated **latent TGF- β (esp. TGF- β 3 in palate; TGF- β 1 in immune cells)** on the cell surface. (*Demonstrated in mouse MEE: GARP indispensable for surface latent TGF β 3.*)
3. Absent surface latent TGF- β → **prevents** integrin ($\alpha\beta$ 8/ $\alpha\beta$ 6)-mediated activation and local presentation of mature TGF- β . (*Demonstrated/mechanistically established.*)
4. Reduced local TGF- β activity → **leads to** decreased **TGF- β receptor engagement and SMAD2/3 phosphorylation**. (*Demonstrated: reduced pSMAD2 in E14.5 palatal MEE.*) 5a. Reduced SMAD2 signaling in palatal MEE → **causes** failed programmed cell death (apoptosis) of the medial edge epithelium → **prevents** palatal shelf fusion → **cleft palate**. (*Demonstrated; phenocopies *Tgfb3*-null.*) 5b. Reduced TGF- β /SMAD signaling in the developing retina → **inferred to** disrupt retinal vascular/neuroepithelial development → **proliferative retinopathy**. (*Inferred, not directly demonstrated.*) 5c. Reduced TGF- β /SMAD signaling in the developing CNS → **inferred to** impair neurodevelopment → **developmental delay** ($\pm$ growth retardation/dysmorphism). (*Inferred.*)

Molecular pathways. Core pathway is **TGF- β /SMAD2-3 signaling** (KEGG hsa04350; Reactome "Signaling by TGF-beta Receptor Complex"). GARP (*LRRC32*) functions as the latent-TGF- β docking receptor upstream of receptor activation. Integrin $\alpha\beta$ 8-mediated activation is the proximal activation step.

Cellular processes. **Apoptosis** of the palatal medial edge epithelium is the key demonstrated cellular event (GO:0006915 apoptotic process; specifically epithelial cell apoptosis during palate fusion). Loss of GARP → reduced MEE apoptosis → persistent midline epithelial seam → cleft.

Protein dysfunction. GARP is a leucine-rich-repeat transmembrane protein; the p.Arg544Ter truncation removes C-terminal/transmembrane-proximal sequence required for surface presentation, and the conserved Ile→Thr missense is predicted to impair folding/function — both yielding **loss of function** (failure to chaperone/present latent TGF- β).

Immune system involvement. GARP is the principal presenter of latent TGF- β 1 on **regulatory T cells and platelets**; in cancer/immunology contexts it drives immunosuppression. In this developmental syndrome the immune role is secondary, but the same biochemistry (GARP:latent-TGF- β :integrin activation) underlies the disease. No overt immunodeficiency/autoimmunity has been reported in patients.

Suggested ontology terms. GO:0007179 (transforming growth factor beta receptor signaling pathway), GO:0006915 (apoptotic process), GO:0060021 (palate development), GO:0001654 (eye development). CL terms: epithelial cell **CL:0000066** (palatal medial edge epithelium), retinal pigment epithelial cell **CL:0002586**, neuron **CL:0000540**. UBERON: secondary palate **UBERON:0001716**, retina **UBERON:0000966**, brain **UBERON:0000955**. CHEBI: TGF- β is a protein (not a small molecule), so no CHEBI term applies to the ligand itself.

7. Anatomical Structures Affected

- **Primary organs:** secondary **palate** (UBERON:0001716), **retina/eye** (UBERON:0000966 retina; UBERON:0000970 eye), **brain/CNS** (UBERON:0000955).
- **Body systems:** craniofacial/digestive-respiratory interface (palate), visual system, central nervous system; with growth (in the 2022 patient).
- **Tissue/cell level:** **palatal medial edge epithelium** (epithelial tissue), **retinal cells** (including retinal pigment epithelium and vasculature), **neural cells**.
- **Subcellular:** GARP is a **plasma-membrane** protein (GO:0005886 plasma membrane); the functional defect is at the **cell surface** (loss of surface latent-TGF- β complex).
- **Localization / lateralization:** cleft palate is typically **midline**; retinopathy and developmental delay are **bilateral**. Specific laterality data are limited given the small cohort.

8. Temporal Development

- **Onset:** **congenital** for the triad; growth retardation is **prenatal-onset** in the 2022 patient.
- **Onset pattern:** developmental/structural (present at birth), not acute.
- **Progression:** the malformation syndrome is largely **static**; however, proliferative retinopathy can **progress** to retinal detachment/vision loss without ophthalmologic intervention.
- **Disease duration:** **chronic, lifelong**.
- **Critical periods:** the palatal defect originates during **secondary palate fusion (~E14.5 in mouse; ~weeks 8–12 of human gestation)** — a window inaccessible to postnatal

intervention. Postnatal critical windows exist for **retinal intervention** and **early developmental/rehabilitative therapy**.

9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic *LRRC32* LoF); both families consanguineous.
- **Penetrance:** appears **complete** in reported homozygotes (all biallelic individuals affected), though based on very few cases.
- **Expressivity: variable** — the 2022 patient showed additional growth retardation/dysmorphism beyond the core triad.
- **Consanguinity:** central to disease occurrence.
- **Carrier frequency:** ultra-low; the recurrent p.(Arg544Ter) allele has gnomAD AF $\approx 1.37 \times 10^{-6}$ with **no homozygotes** — consistent with a severe, ultra-rare recessive condition.
- **Founder effects / anticipation / mosaicism:** No founder effect, genetic anticipation, or germline mosaicism has been documented. Anticipation is not expected (not a repeat-expansion disorder).
- **Epidemiology: prevalence and incidence are undefined** — fewer than ~5 patients reported worldwide. **Sex ratio, geographic distribution, and age distribution** cannot be reliably estimated.

10. Diagnostics

- **Recommended approach:** molecular genetic testing. **Whole-exome sequencing (WES)** was the diagnostic modality in both the original and second reports and is the highest-yield test; **whole-genome sequencing (WGS)** is a reasonable alternative. Targeted single-gene testing of *LRRC32* is appropriate once the phenotype is recognized or a familial variant is known.
- **Variant interpretation:** confirm biallelic status; classify per **ACMG/AMP** with ClinVar/segregation support.
- **Clinical/imaging workup:** ophthalmologic examination and retinal imaging (to characterize the proliferative retinopathy), craniofacial/palate examination, and developmental assessment. **Chromosomal microarray/karyotype** are typically normal (point-mutation disorder) but may be performed to exclude structural mimics.
- **Differential diagnosis:** other syndromic cleft-palate disorders (e.g., *TGFB3*-related, Van der Woude, **Stickler syndrome** — which also features cleft palate plus retinal detachment and

is thus an important mimic), and other syndromic retinopathy/developmental-delay conditions. Molecular testing distinguishes them.

- **Screening:** for at-risk consanguineous families with a known familial variant, **carrier testing** and **cascade/prenatal testing** are available. No population newborn screening exists.

11. Outcome / Prognosis

Formal survival, mortality, and quality-of-life outcome data are **not available** given the rarity. Inferences: the human disorder is compatible with **postnatal survival** (unlike the neonatal-lethal complete mouse null, human patients survive infancy — likely reflecting residual/hypomorphic function or species differences). Prognosis is shaped by the **severity of the retinopathy** (vision-threatening) and the **degree of developmental delay** (affecting long-term function), plus feeding/speech consequences of the cleft palate. Recovery of the structural defects is not spontaneous; surgical/rehabilitative intervention improves function. Prognostic biomarkers are **not established**.

12. Treatment

No targeted or disease-modifying therapy exists. Management is **supportive and multidisciplinary**:

Domain	Intervention	Suggested NCIT concept
Cleft palate	Palatoplasty (surgical repair); feeding support; speech therapy	Cleft palate repair; Speech therapy
Proliferative retinopathy	Ophthalmologic surveillance; vitreoretinal surgery ; consider anti-VEGF/laser per lesion type	Vitrectomy; Laser therapy
Developmental delay	Early developmental intervention ; physical/occupational/speech therapy; special education	Rehabilitation therapy
Family	Genetic counseling (25% sibling recurrence)	Genetic counseling

Pharmacotherapy / pharmacogenomics / gene therapy / cell therapy / RNA therapy / immunotherapy: None approved or in trials for this disorder specifically. No NCT-registered trials target MONDO:0033641. (Note: the broader GARP-TGF- β axis is an active *oncology* immunotherapy target — anti-GARP antibodies, CAR-T, bispecifics — but these aim to *inhibit* GARP and are unrelated to treating GARP-deficiency syndrome.)

Experimental outlook: As a monogenic LoF disorder, it is conceptually a candidate for future gene-replacement approaches, but the congenital/developmental timing of the palatal defect limits postnatal correction of already-formed malformations.

13. Prevention

- **Primary prevention:** none at the population level. For known-carrier couples, **preimplantation genetic testing (PGT)** and **prenatal diagnosis** can prevent affected births.
- **Secondary prevention:** **cascade carrier testing** in consanguineous families with a known variant; early ophthalmologic screening in an at-risk newborn to catch progressive retinopathy.
- **Tertiary prevention:** timely palatoplasty, retinal intervention, and developmental therapy to prevent complications (malnutrition, aspiration, vision loss, functional decline).
- **Counseling:** **genetic counseling** is central — autosomal-recessive 25% recurrence risk; discussion of consanguinity.
- **Immunization / public health / environmental interventions:** not applicable.

14. Other Species / Natural Disease

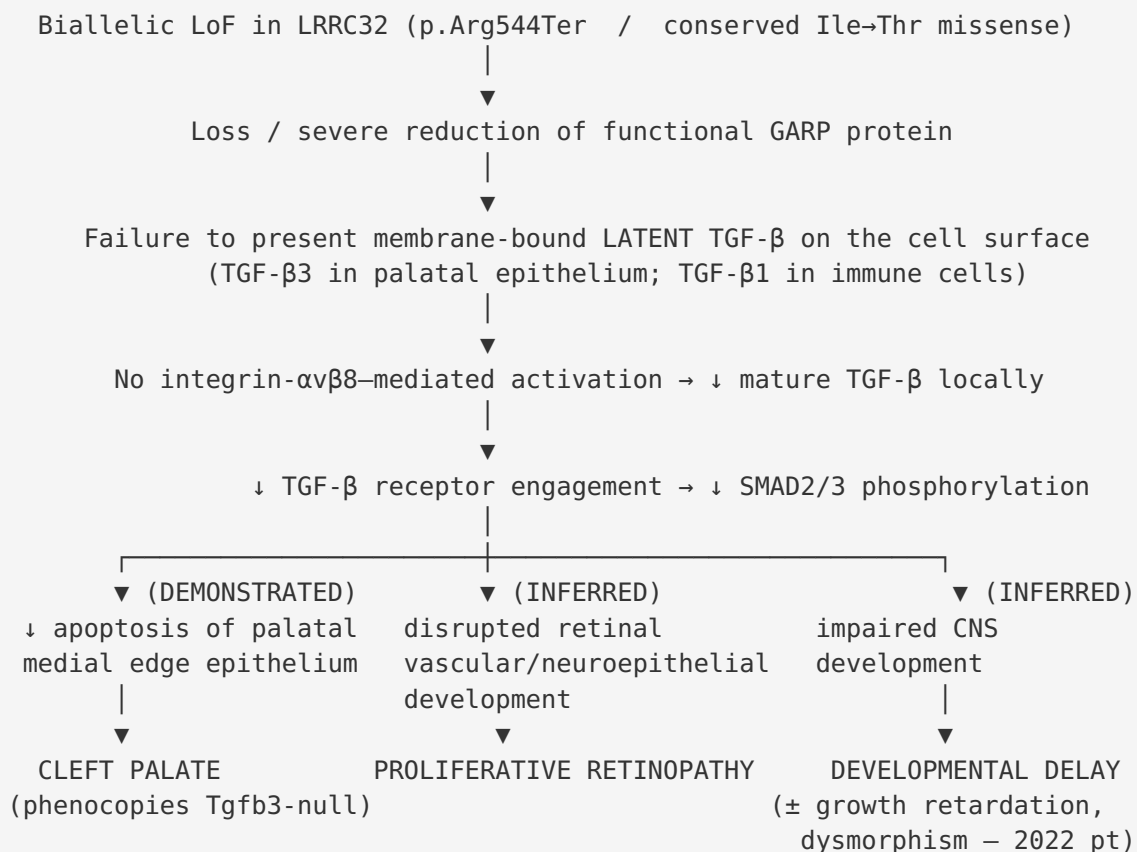
- **Model species:** *Mus musculus* (NCBI Taxon:10090) — *Garp/Lrrc32* knockout. No naturally occurring animal disease (OMIA) is documented for this syndrome.
- **Orthologous gene:** mouse **Lrrc32/Garp**; the gene and its TGF- β -presenting function are evolutionarily conserved.
- **Comparative pathology:** the mouse null reproduces the **cleft palate** but is **neonatal-lethal**, so it does not model retinopathy or developmental delay. **Zoonotic/cross-species transmission:** not applicable (genetic disorder).

15. Model Organisms

- **Model type / system:** mammalian — **mouse *Garp/Lrrc32* knockout** (constitutive null). In vitro/ex vivo palatal shelf cultures and MEE cell assays supplement the model.

- **Genetic models available:** constitutive knockout demonstrated; conditional/tissue-specific alleles would be needed to bypass neonatal lethality (see follow-up).
- **Phenotype recapitulation: Good for the palatal phenotype** — the KO shows isolated secondary palate fusion failure with reduced MEE apoptosis and pSMAD2, phenocopying *Tgfb3*-null mice. **Poor for ocular/neurodevelopmental phenotypes** — death within 24 h of birth precludes their assessment.
- **Applications:** dissecting GARP → latent-TGF-β3 → SMAD2 → MEE-apoptosis axis in palatogenesis; testing whether TGF-β pathway restoration rescues palate fusion.
- **Resources:** MGI (mouse), IMPC/KOMP for allele availability.

Mechanistic Model / Interpretation



The **upstream, demonstrated** portion of this chain (mutation → loss of GARP → loss of surface latent TGF- β → ↓ SMAD2 → ↓ MEE apoptosis → cleft palate) rests on direct experimental evidence in the mouse. The **downstream retinal and neurodevelopmental branches are inferred** from (a) the established requirement of TGF- β signaling for retinal and CNS development and (b) the co-segregation of these features with the biallelic *LRRC32* genotype in humans. The consistency of a single, well-understood signaling deficit across all three affected tissues makes the unified GARP–TGF- β model parsimonious and compelling, while honestly flagging that the ocular and neural mechanisms have not been directly proven in this disease.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
30976112	<i>Homozygous stop-gain variant in LRRC32... cleft palate, proliferative retinopathy, developmental delay</i> (Harel et al. 2019)	Human clinical/genetic	Foundational — establishes gene–disease link, triad, TGF- β rationale, mouse lethality
35656379	<i>A Novel Homozygous Missense Variant in LRRC32</i> (Hexner-Erichman et al. 2022)	Human clinical/genetic	Confirmatory — 2nd family, allelic heterogeneity, phenotype expansion
28912269	<i>GARP positively regulates TGFβ3 and is essential for mouse palatogenesis</i> (Wu et al. 2017)	Model organism/in vitro	Mechanistic core — GARP→latent-TGF β 3→SMAD2→MEE apoptosis; phenocopies <i>Tgfb3</i> -null
gnomAD v4	<i>LRRC32</i> constraint & allele frequencies	Computational/population	Recessive-compatible constraint; ultra-rare allele, no homozygotes

Supporting context (GARP–TGF- β axis biology). Multiple recent structural and immunological studies corroborate that GARP presents latent TGF- β on the cell surface and that activation is integrin-dependent — e.g., cryo-EM/allostery work showing α v β 8-mediated activation of L-TGF- β 1/GARP ([PMID: 39288764](#)), and reviews/therapeutic studies confirming GARP (*LRRC32*) as the docking receptor presenting latent TGF- β on Tregs and platelets ([PMID:](#)

[35898500](#); [PMID: 36928178](#)). These strengthen the biochemical plausibility of the disease mechanism, although they address immuno-oncology rather than the developmental syndrome directly.

Challenges / caveats to the model. Newer "dynamic allostery" data suggest mature TGF- β may signal *without physical release* from the latent complex ([PMID: 39288764](#)); this refines but does not overturn the conclusion that GARP loss impairs TGF- β activation. The extensive PVR (proliferative vitreoretinopathy) literature involving TGF- β -driven RPE epithelial–mesenchymal transition ([PMID: 42285191](#); [PMID: 40466854](#)) concerns *acquired adult* retinopathy and is mechanistically distinct from the *congenital* retinopathy in this syndrome — a reminder that "proliferative retinopathy" here should not be conflated with adult PVR.

Limitations and Knowledge Gaps

1. **Extreme rarity (~4 patients).** All clinical inferences rest on a handful of individuals from consanguineous families; prevalence, incidence, sex ratio, penetrance, expressivity, and natural history are essentially undefined.
 2. **Ocular and neurodevelopmental mechanisms are inferred, not demonstrated.** The mouse null's neonatal lethality prevents modeling of retinopathy and developmental delay; no tissue-specific evidence links GARP loss to these features directly.
 3. **No confirmed OMIM phenotype / Orphanet / ICD unified code** was identified for the triad syndrome, complicating standardized annotation.
 4. **No treatment evidence base.** Management is extrapolated from component-condition standards; no outcome, response-rate, or QoL data specific to this syndrome exist.
 5. **Genotype–phenotype correlation is unresolved.** Whether the missense (2022) vs. nonsense (2019) alleles produce systematically different severity (e.g., the added growth retardation) cannot be determined from two families.
 6. **Human residual function unexplained.** Why humans survive infancy while complete mouse nulls die within 24 h is unknown (hypomorphic alleles? species differences in GARP dependence?).
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Proposed Follow-up Experiments / Actions

1. **Conditional/tissue-specific mouse models** (e.g., retina- and CNS-specific *Lrrc32* knockouts, or hypomorphic/knock-in alleles mimicking the human p.Arg544Ter and Ile→Thr variants) to bypass neonatal lethality and **directly test the retinal and neurodevelopmental branches** of the causal chain.
2. **Functional validation of the missense allele** — express the conserved Ile→Thr variant in cells and assay surface presentation of latent TGF- β and SMAD2 activation to confirm loss-of-function and quantify residual activity relative to the nonsense allele.
3. **Patient registry / GeneMatcher outreach** to identify additional families, enabling genotype–phenotype correlation, penetrance/expressivity estimates, and natural-history documentation.
4. **iPSC-derived retinal organoids and cortical organoids** from patient cells (or CRISPR-engineered *LRRC32* nulls) to model the human-specific ocular and neural phenotypes in vitro and probe TGF- β /SMAD signaling deficits.
5. **Formal variant curation** in ClinVar/ClinGen and assignment/confirmation of OMIM and Orphanet identifiers to standardize the disease entry.
6. **Ophthalmologic natural-history study** in any identified patients to define the trajectory of the proliferative retinopathy and optimal intervention windows.

Report compiled from a five-iteration autonomous investigation. Evidence sources are distinguished as human clinical

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and computational/population-genetic (gnomAD v4). Findings F001–F004 are recorded in the knowledge state with verified abstract quotations.